

AI Fundamentals: Use AI with Judgment

A self-paced online certificate course · 5 modules · about 20 hours

Executive summary for future participants

AI helps on some tasks and hurts on others

In a field experiment with 758 consultants using GPT-4, the same tool raised quality on some tasks and lowered accuracy on a task just beyond its reach. The boundary was invisible to the people using it.

The skill this course teaches: knowing which kind of task you are facing, and checking before you trust.

40%+

higher-quality work on
tasks inside AI's
capability frontier

-19 pts

in correct answers on a
task just outside it

Source: Dell'Acqua et al., "Navigating the Jagged Technological Frontier,"
Organization Science (2026); working paper 2023.

Five modules, from how AI works to how you use it



Module 1

How Modern AI Works

What kind of thing is this?



Module 2

AI in Science and Discovery

How can we tell good AI-based science from bad?



Module 3

Working with AI

How should I divide work between myself and AI?



Module 4

AI and Society

Who benefits, who bears the costs, and how do we know?



Module 5

AI in Personal Life + Capstone

What kind of relationship do I want with AI?

About 20 hours in total

Self-paced, fully online

About 4 hours per module

Certificate of completion

What you will be able to do, and why it matters

1

Explain how AI evolved from rules to foundation models

Why it matters: Calibrate how much to trust AI on a given task

2

Evaluate AI-supported claims using evidence tiers

Why it matters: Separate real breakthroughs from hype

3

Collaborate with and delegate to AI, including agents

Why it matters: Gain speed without losing control

4

Analyze AI's social, economic, informational and environmental effects

Why it matters: See costs that others may bear

5

Design your own Personal AI Use Charter

Why it matters: Turn what you learn into a working plan

6

Apply the SEATS critical-thinking check

Why it matters: Keep a habit that outlasts any tool

SEATS: five questions to ask before you trust AI



Source

Where does this
come from?



Evidence

How strong is the
support for it?



Alternatives

What else could
explain it?



Test

Can I check it
myself?



Stakes

What happens if it
is wrong?

Practiced in every module, not just taught once. You rate yourself on each step in your first journal entry and again at the end, so you can see your own growth. The higher the stakes, the more steps you run.

Learn by doing, with the AI tools you already use



M1 Predict the Failure

Guess where AI will fail on 10 tasks, then test your guesses



M2 Audit an AI Science Claim

Trace a headline back to its original evidence



M3 Delegation Drill + Personal Eval Set

Supervise a multi-step task; build a test of AI on your own work



M4 Provenance and Footprint Lab

Verify the origin of images; estimate the energy of your AI use



M5 Sycophancy Probe + Privacy Audit

See whether AI just agrees with you; review what your tools keep

Three threads run through all five modules

AI Field Journal

10 short reflections on your real AI use

Personal Eval Set

6 test items from your own work, re-run as tools change

Personal AI Use Charter

Drafted along the way, completed as your capstone

The hidden cost: cognitive offloading

Letting a tool do our mental work saves effort, but it can also remove the practice that keeps our skills sharp. Early evidence from three different settings:

28% → 22%

Endoscopists' detection rate in procedures without AI, after routine AI use began

Budzyń et al., Lancet Gastroenterol. Hepatol. (2025), observational

Gains vanished

Students with an unrestricted AI tutor improved in practice, then did worse once AI was removed; a hint-giving tutor largely avoided the harm

Bastani et al., PNAS (2025), field experiment

319 workers

Knowledge workers who trusted AI more reported less critical thinking in their AI-assisted tasks

Lee et al., CHI (2025), survey



Treat these as early warning signals, not settled conclusions: the studies differ in design, setting and size.

Use AI to grow your skills, not replace them



Attempt first

Draft, solve or outline on your own before asking AI



Ask for hints, not answers

Use AI as a tutor or critic when you are learning



Practice key skills without AI

Schedule regular AI-free work on what matters most



Match checking to stakes

Run the full SEATS check when errors are costly



Re-test as tools change

Re-run your own eval set; don't rely on benchmarks



Decide what stays human

Keep some judgments, relationships and decisions yours

Built into Module 3 (Protecting Your Expertise), Module 5 (Learning with AI) and the skill-protection section of your charter.

Your capstone: a Personal AI Use Charter

Map your own tasks into four zones

Delegate

Low stakes, easy to check

Collaborate

AI drafts or critiques; you decide

Verify first

Useful, but errors are costly

Keep human

Judgment, sensitive data, skills to protect

How it works

1

Draft along the way

Journal entries in Modules 3 and 4

2

Write 800–1,000 words

Cite course evidence and your eval results

3

Peer review

Give and receive two rubric-based reviews

4

Keep it

Download it; revisit it as tools change

Use AI. Keep your judgment.

Tools change within months. The skills of evaluating, delegating and verifying do not.



Certificate of
completion



Your Personal AI
Use Charter



Your own AI
evaluation set



The SEATS thinking
habit